

Tae Yong Song

Ph.D. Student in Building Science

Department of Architecture and Architectural Engineering, Seoul National University

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RESEARCH PROFILE

I study building intelligence at the intersection of building physics, performance simulation, and artificial intelligence. My work focuses on models that remain physically meaningful, transferable, and useful for building-energy decisions when operational data are limited.

EDUCATION

2026–present	Ph.D. in Architecture and Architectural Engineering , Seoul National University Building science and building performance simulation
2022–2023	M.S. in Engineering , Seoul National University Thesis: “Exploring Potential of Transfer Learning in Overcoming Data Scarcity for Building Energy Model”
2015–2021	B.S. in Architecture and Architectural Engineering , Seoul National University Graduated in 2021

RESEARCH INTERESTS

- Physics-embedded and physics-informed machine learning
- Building energy and thermal performance
- Building performance simulation and reduced-order modeling
- Data-efficient learning, transfer learning, and digital twins

PUBLICATIONS

International Journal Article

- 2026** Physics-embedded hybrid modelling approach for room temperature prediction using Siamese neural network and RC model

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, Junu Lee, and Cheol-Soo Park. *Journal of Building Performance Simulation*, 19(2), 277–288.

International Conference Papers

- 2026** Curated Intelligence for Visual Inference and Deterministic Logic toward Automatic BEM

Song, T. Y., Kim, J. H., Lee, S. J., Kim, S. K., Park, H. C., Kim, K. J., and Park, C. S. *Proceedings of the 7th Building Simulation Applications Conference*, June 24–26, Bozen-Bolzano, Italy.
- 2024** Physics-Informed Hybrid Modeling Approach for Room Temperature Prediction Using an RC Model and Siamese Neural Network

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, and Cheol-Soo Park. *SimBuild 2024*.
- 2023** Causality-based model for decision-making of building energy retrofit

Chul-Hong Park, Young-Sub Kim, **Tae Yong Song**, Seon-Jung Ra, and Cheol-Soo Park. *Building Simulation 2023*, 1433–1438.
- 2023** Limitations of Conventional Artificial Neural Network-Based Surrogate Models for Building Energy Retrofit

Chul-Hong Park, Young-Sub Kim, Seon-Jung Ra, **Tae Yong Song**, and Cheol-Soo Park. *ASHRAE Annual Conference*, 683–690.

Domestic Conference Papers

- 2026** Development of Customized AI Technologies for BIM-to-BEM Automation: Integration of LLMs, Computer Vision, Delaunay Geometry Algorithms, and Minimum Cycle Algorithms
Tae Yong Song, Jin-Hong Kim, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 2026 Spring Annual Conference of the Architectural Institute of Korea*, 46(1), 829–830. **Awarded the Outstanding Presentation Paper Award (General Division, Oral Presentation).**
- 2022** Dynamic Sensitivity Analysis Using NODE-GAM
Tae Yong Song, Chul-Hong Park, Seon-Jung Ra, Young-Sub Kim, and Cheol-Soo Park. *Proceedings of the 2022 Annual Conference of the Korean Institute of Architectural Sustainable Environment and Building Systems*, 181–182.

RESEARCH PROJECTS

- 2026–2030** **Virtual Building Modeling and Simulation Framework, and a Causality-Based AR Tool for Net-Zero Building Design Evaluation**
Korea Institute of Civil Engineering and Building Technology (KICT)
Contributing to a virtual-building simulation framework, causal performance evaluation, and AR-based communication of net-zero design outcomes.
- 2025–2026** **LLM Technology for Digital-Twin-Based Building Energy Models**
Samsung Electronics
Developed LLM-assisted workflows for organizing, generating, and interpreting building-energy model information in a digital-twin environment.
- 2025–2026** **IDF Model Development for Digital-Twin Energy Simulation**
Eterion (Samsung Electronics)
Developed and reviewed simulation-ready EnergyPlus IDF models and model-validation workflows for building-energy digital twins.
- 2022–2024** **Automated Digital Diagnosis and Design Technologies for Green Remodeling**
Korea Agency for Infrastructure Technology Advancement (KAIA)
Contributed to building-energy modeling, data-driven performance diagnosis, and automated evaluation of green-remodeling alternatives.
- 2023** **Data-Driven HVAC Load Prediction and Optimal Air-Handling-Unit Control**
LG Electronics
Developed and evaluated HVAC load-prediction models for control-oriented air-handling-unit operation.
- 2022–2023** **Greenhouse-Gas Reduction Factors and Emissions Assessment for Building Green Remodeling**
Ministry of Land, Infrastructure and Transport (MOLIT)
Supported energy simulation and scenario analysis for estimating greenhouse-gas reduction factors under alternative retrofit conditions.

TEACHING EXPERIENCE

- 2026–present** **Teaching Assistant**, Seoul National University
Department of Architecture and Architectural Engineering
- 2023** **Teaching Assistant**, Seoul National University
Department of Architecture and Architectural Engineering

HONOR

- 2026** **Outstanding Presentation Paper Award**
Architectural Institute of Korea Spring Annual Conference, Oral Presentation – General Division